Guaman Poma and *Quipo* Cryptography: a suggested reading of Andean knotted cords

> "It is a peculiarity of our epoch that, at the moment when the phoneticization of writing ... begins to lay hold on world culture, science, in its advancements, can no longer be satisfied with it ... Within cultures practicing so-called phonetic production, mathematics is not just an enclave ... This enclave is also the place where the practice of scientific language challenges intrinsically and within increasingly profundity the ideal of phonetic writing ... But beyond theoretical mathematics, the development of the practical methods of information retrieval extends the possibilities of the message vastly, to the point where it is no longer the written translation of a language"[1] (Derrida 9-10).

Unlike the notable civilizations of Mesoamerica, the Inca empire of Peru had been conquered and administered without the benefit of phonetic writing. Instead, an elaborate system of knotted cords called *quipos,* kept detailed accounts and records for the succession of ruling elites. Despite their importance to Inca society, *quipos* have been excluded from the larger family of writing systems. Any Peruvianist scholars see them as arithmetic records, devoid of any linguistic references; therefore, most scholarship uses a mathematical perspective for any analysis of their function.[2] This approach has often led to a misunderstanding of what *quipos* can signify. In some cases, working ideas fail to take account of fundamental ancient Andean number theory. Others presuppose a simplistic model for the *quipo* that strip these mnemonic devices of much of their meaning.

The aim of this paper is to clarify problems created by previous approaches to *quipos*. I suggest that a new model for understanding *quipos* should be built first on a close reading of colonial sources, especially those of the native chronicler Guaman Poma de Ayala. Here connections between the role both *quipos* and *quipocamayos*, the keepers of these knotted cords, are played out in a dynamic Andean landscape. On this basis, a critique of past mathematical work is included to better represent how *quipos* actually worked. Models then maybe based more on functional operation, rather than on mathematics. Examples raised by Guaman Poma's descriptions and drawings suggest, in fact that *quipos* could function as memory markers, in regards to knot placement as well as color. Other examples, found in Inca textiles, may in fact show *quipos* have a color specific iconography. In tum I suggest new directions in which scholarship may proceed in attempting to decipher this enigmatic code.

---

[1] Taken from Elizabeth Boone's essay in Writing without Words.

[2] For a further discussion on mathematics see Gary Urton "Messages from the Mallquis," unpublished paper.

Spanish Historicity and *Quipos*

Analysis typically begins with the assumption that mnemonic knots and cords are "clumsy" as compared to phonetized writing. Knots are considered both a predecessor to hieroglyphic representation and the mark of tribal culture. In her book *Quipus* and Witches Knots, Cyrus Lawrence Day enumerates the two chief functions of mnemonic knots as utilized by "primitive man": numerical, to record dates and numbers; and cultural, to help preserve the memory of songs, stories, genealogies, historical traditions, and religious laws. Examples from the history of man's use in knotted records include the Peruvian example. Her argument and western anthropological bias qualify knotted cords as, "not by their very nature, convenient device" to record information (Day 13). Knots, in her view are a cumbersome method of record keeping, especially when manipulation and handling are concerned.

This model is in stark contrast to Spanish historical narrative evidence where *quipos* are concerned. Spanish and native chroniclers of the colonial period describe these knotted cords and their *quipocamayos* as essential to Inca imperial administration. The Inca "had a very cunning method of counting wool and knots, the wool being of different colours. They call them *quipus* and they are able to understand so much by their means, that they can give account of all the events that have happened in their land for more than five hundred years" (Molina 10). Sixteenth-century writers recorded the testimonies of many *quipocamayos* for information on Andean history. For chronicler Cieza de Leon, the size and expanse of tasks throughout the Empire necessitated a complex communicative administrative system (Ascher 8-11). Different individuals, corporate groups, and communities retained these learned men to settle disputes in Spanish courts (Markham 8).

Quipocamayos held a position multifaceted in character. They furnished the government with information on various matters. "One had charge of revenues, reported the quantity of raw materials, distributed among labor, the quantity and quality of the fabrics made from it, and the number of stores, of various kinds, paid in royal magazines" (Prescott 134).[3] Others detailed references for a yearly tribute census in distant provinces. Cieza De Leon writes that the outputs of gold mines, the compositions of work forces, the amount and kinds of tribute, the contents of storehouses, "down to the last sandal" were all recorded on *quipos* (Ascher 10). These returns were forwarded to the capital of Cuzco, where they submitted them to other officers for inspection, all of whom were acquainted with the art of reading *quipo* code. Other early chroniclers describe the functionality of their social position, recording accounts of other kinds of *quipocamayos*. Some were analysts, entrusted with remembering the important events that occurred in their stationed province or during the reign of each Inca dynasty. Others were troubadours, who performed ballads and recited poems at various public gatherings; still others kept records of the eclipses of the sun, phases of the moon, and the movement of the stars (Muna and Moms 23; Prescott 135-136; Rowe 1945, 1946: 201,326).

Guaman Poma as quipocamayo

It is very doubtful that Day's argument of "primitive" culture considered such complex models of association. In the Inca Empire the *quipocamayos* served as a mixture

[3] Recent scholars referencing Guaman Poma now accept Prescott's early work.

of historians, accountants, and astrologists. Their importance to rulership and Inca hegemony placed them in an elite rank of the hierarchy. For at least one native colonial chronicler, Guaman Poma de Ayala, this privilege allowed him to learn to read and write Spanish, ultimately to record his own view of colonial matters in the form of a letter. Work presented by historian Juan M. Ossio has shed much light on Guaman Poma's relationship to pre-colonial *quipocamayos*. He writes

> "que todas los nombres de los personajes con *quipos* de la Nueva Cor6nica figuran en una red familiar que el describi6 al hablar de la cuarta edad del mundo, o Aucaruna" and that "su disposicion para desempefiarse como interprete y aprender a leer y cribirse se debia a que decendia de una familia de lectores de *quipos* // that all of the personalities with *quipos* in the Nueva Coronica belong to family lineage that was described in the forth era of the world, or Aucaruna" and that "his disposition to place interpretation and learning and reading is due to belonging to a family of *quipo* readers" (Ossio 47).[4]

Much of the actual document, *El Premier Nueva Cor6nica y Buen Gobierno* can be read as an example of this connection. Ossio suggests that it would be impossible to fathom the production of a 1200-page letter to the king of Spain in colonial Peru, without the compiler himself being a *quipocamayo*.

Guaman Poma's choice and effort to create the *Nueva Cor6nica* can also be understood as a continuation of this family tradition. In this medium he not only tries to keep oral and *quipo* traditions alive, but also argues for a reassessment of the Spanish regime.

> "Al llamarse cronista, Guaman Poma rebanaba una prerrogativa que conciencia el bien publico y se mantearia por encima del interes personal. El se habilito pan la labor del historiador al proclamar su devoci6n a la religion cristiana y al forma que su historia realizaria los servicios de toda buen histor. // Calling himself chronicler, Guaman Poma develops a prerogative dedicated to the public good and it maintained itself above any personal interests. He justified historical work by proclaiming his devotion to Christian religion and a form that his history realized to service of all good history" (Adorno 1989:23).

Guaman Poma and other later indigenous chroniclers who began to write in a foreign dominant culture found themselves defending and reconstructing their own identity in this new paradigm. "The search for adjustment to various models [of Spanish hegemony] belies the multiple subject positions [chronicler, *quipocamayo*, Christian] that the colonial subject seemed destined to take, not merely sequentially, but most often simultaneously" (Adorno 1991, 258). Guaman Poma argues for the segregation of the Andean and Spanish realm and at same time develops a thread of legitimacy for his own lineage. Writing was not a tool for the legitimization of the Spanish regime, but a new way to describe an anti-colonial stance. The *quipos* themselves form a structural element in this argument, mediating the authority of Inca governing.

---

[4] 4 This any subsequent translations are my own.

Quipo as mnemonic device and astrological tools in the *Nueva Coronica*

Drawing on his multiple roles, Guaman Poma uses *quipocamayos* and *quipo* imagery to explain the structure once present in Inca imperial administrations. In the *Nueva Cor6nica*, there are various references to *quipos* and *quipocamayocs* and seven illustrations of *quipos* about his text. "Guaman Poma's drawings ... illustrate an integrated system of time-space that he never explicitly mentioned in his text; he serves to confirm and correct the information of other written sources of which he may not have been aware" (Zuidema 163). He makes various explicit references to the use and function of the *quipo*, not only for records and numeration, but also as a mnemonic device that allowed for the smooth transfer of Andean oral traditions. His first treatment of *quipo* in the text references this connection between oral and written traditions in pre-colonial times (Guaman Poma 8-11). One of his first *quipo* illustrations also suggests this mnemonic connection. In his drawing of a messenger boy (Plate 1), or *sayapayac*, he depicts him carrying a small *quipo*. Guaman Poma labels it not by its *Quecha* name, but instead uses the Spanish word *carta*, to denote that this *quipo* was a letter (202). This suggests that *quipos* not only carried information for a prospective reader, but also were the Andean version of a message, or written information. One could also argue that the label *carta* indicates for the intended audience (ie. the king) what the image is intended to convey. The image cannot do this by itself; hence this gloss performs multiple translative tasks.

Later on in his letter, Guaman Poma depicts the *quipo* and the mnemonic signs as they are integrated into the colonial landscape. He makes two mentions of the use of *quipos* as mnemonic rosaries. First as "de limonsa para que sean cristianos/ for penitence so that they maybe Christians" (652), second for the confessions of sins, as "*quipo* de sus pecados // *quipo* of their sins" (616). Both are indications that *quipos* were used even after the conquest. Later in the text he makes an argument for the readers of *quipocamayos*, in the "Consideracion," as not only learned, but good Christians (888). Here the aim is a defense of tradition in an age of Spanish colonialism, even though the *quipo* for the early part of Spanish colonialism was never under attack. "The incommensurability between the form of the *quipo* and European forms of memory meant that the *quipo* was not immediately the object of early campaigns of destruction as were [codices] of Mexico that were burned" (Cummins 194). *Quipos* are understood in the *Nueva Cor6nica* to hold their own with the Spanish written word, as a window into the very heart of Andean understanding.

These ideas are better developed not in Christian dogma, but in Guaman Poma's treatment of *quipos* as astrological markers. Peruvians had accurate knowledge of the movements of the sun, the moon, and of the planets Mercury, Venus and Jupiter. "They had observation towers, four east of Cuzco, and four to the west, for observations" (Day 27). Their purpose in observing and calculating solstices and equinoxes was to regulate the calendar and to fix the proper times to sow and harvest. This denotes a high degree of complexity in mathematics and calculation, all made possible using *quipo* notation. On page 883 of the Nueva Cor6nica, there appears an illustration of such an astrologer *quipocamayo* with his *quipo* in hand (Plate 2). In the text, Guaman Poma explains the *quipo's* purpose within divination and astrology. One can come to understand the *quipo*

image as a mnemonic device far more complex than mere records of numbers and calculations. As demonstrated by the *Nueva Cor6nica*, *quipos* in pre-colonial use served various functions and were coded with different readings depending on subject matter. Often passed scholarship fails to recognize the layers of meaning possible in their use.

Quipo counts: Locke and Nordenskiold

Most past scholarship on *quipo* decipherment is inadequate. Much scholarship was conducted to foster a better understanding of these knotted cords. Treatments often begin with the astrological notion ascribed to *quipos* and their calculations. However, they tend to ignore that *quipos* worked on many levels beyond numeric calculation and displacement. First among these *quipo* theoreticians are Leland Locke and Briand Nordenskiold, who worked separately but used the same schematic to count knots on the pendants of extant *quipo* cords. Simply said, their model was counting the number of knots on a particular pendant cord. Placement in this model denotes quantity. In Andean mathematics, because of the use of decimal notation, pendant strings are understood to represent tenth places in descending order down a particular cord (Plate 3). All the specimens examined were grave *quipos*, connected to religious divination and astrological significance (Locke 4). Close attention was paid to numeration that could be linked to cosmic correlations.

Under their model, the number seven and a propensity for prime numbers were found in grave *quipos*. Prime numbers are those with no other whole numbers as a factor. Nordenskiold suggests such numbers were "magic" and that therefore *quipos* were supplied to the deceased "to occupy himself with, and to prevent him, possibly by means of the magic numbers, from walking after death" (36). After totals were taken of subsequent pendant cords, a numerous quantity off numbers appeared with specific astronomical correlations. Locke suggests that "grave *quipos* were calendrical in nature and may have been used for astronomical or divination purposes" (4). Both agree that *quipos* were primarily used for recording numbers. A secondary purpose was as a memorization tool, but not in any way a conventional scheme for phonetic writing. Locke and Nordenskiold gather this from Spanish colonial chroniclers, yet they do not theorize about a possible connection between numbers in knots and their use as mnemonic devices. For the most part this analysis ignores the very construction of subtle ties, twists, and knots in each cord. No analysis on color is provided either, and like most scholars, they are short of making any attempt at linking cords with specificity for mnemonic use.

Wassen, the Inca Abacus, and List Analysis

Knotted cords by their structure are not computational devices. Knots would clearly take time to tie and even more time to untie, tangle and untangle when manipulated. It is the contention of previous scholarship that *quipo* code was generated on "simple" abacus type instruments. Successive steps therefore could be carried out on boards speedily and accurately. Henry Wassen made this argument in 1931, based on Guaman Poma's illustration on page 360 of the *Nueva Cor6nica*. This drawing depicts the Inca's treasurer holding a quipo larger than all others illustrated in this document. In the left-hand comer of the drawings there is a rectangle divided into twenty squares containing in systematic arrangement several small circles and dots (Plate 4). Wassen writes, "the Indians used to work out computations the result of which were subsequently

recorded by knots on the cords of the quipos" (193).

In support of his conclusion, Wassen quotes Father Jose de Acosta, who lived in Peru from 1571to1586 and published the chronicle *Historia Natural y Moral de las lndias* in 1596:

> "To see them use another kind of *quipu* with maize kernels is perfect joy. In order to effect a very difficult computation for which and able calculator would require pen and ink for various methods of calculation these Indians make use of kernels. They place one here, three somewhere else and eight I know not where. They move one kernel here and three and the fact is that they are able to complete their computations without making the smallest mistake" (Wassen 194).

Wassen mentions the use of pebbles, as well as grains of maize, for mathematical computation. Given this evidence, Wassen uses Guaman Poma's text to explain the vertical axis of the table; a counting description of units of tens, hundreds, and thousands. Yet his analysis of the horizontal axis, which has been followed by other scholars, demands a closer reading:

> "In regard to the numerical values represented by horizontal rows, they are based on the important part that the number five, i.e. a hand, plays among many peoples, evidently also in the past has played in the development of the decimal system in Peru. Thus, in the holes found [plate 4] a sum of three may be placed in symbolism of the number of a certain unit. Should it be required to mark a further three units the square would be overfilled, wherefore one of the holes in the square is at hand to substitute five units of the [next], and so on" (Wassen 193)

Guaman Poma makes no comment on the referential numeration present on the horizontal axis, so how can Wassen make his assumption? His treatment reveals a distinct detachment from the image itself, akin to a straight mathematical exploitation of the text. Wassen also tends to ignore the common use of decimal notation throughout the Andes. Gordon Walker, executive director of the American Mathematical Society, states Wassen's model cannot be accepted. Walker stresses that this system cannot work, especially "unless evidence is presented to substantiate the special role of 5, 15, and 30 in Peruvian arithmetic" (Day 37). The multiples of three Wassen uses are inconsistent with proper decimal positional notation and are distinctly unnatural for conducting calculations.[5]

Another problem with Wassen's claim surfaces in his assumption that the unfilled dots, those not blacked out on the "abacus" board, denote zero. It would be inconsistent with the Andean notation system in *quipos* to assume such a fact. "Only two people, the Indians of lndia and the Indians of Americas (Mayan and Peruvian), are credited with having invented ways to represent the mathematical concept of nothingness" (Day 17). As shown by Locke and Nordenskiold, zero is denoted by the absence of knots from their regular place holding on a *quipo*. *Quipocamayos* would simply skip a step in knotted cords when they wanted to denote zero. Therefore, we may gather that on the table depicted on page 360 of the *Nueva Cor6nica*, the open dots are not zero, but necessarily

---

[5] An excellent source for the study of Andean mathematics and numerology is Gary Urton's *Social Life of Numbers*.

place holders, just like their black opposites. R.Tom Zuidema, in a unpublished lecture, has pointed out that a match can been found of the number of pendant cords in the *quipo* on page 360 and the number of place holders in the same table, totaling 55. This point may also disprove Wassen's idea that white denoted absence and therefore zero. Zuidema assumes that a qualitative correlation can be generated between the fifty-five pendant cords, linking each respective cord with a position on the table. It is therefore difficult to believe that Guaman Poma would choose to represent 33 empty cords on a *quipo*. Extant *quipos* are often found without knots, but never in that frequency. Hector Calederon and Emilio Mendizabal made another suggestion

> "que debido a la necesidad de seftalar las operaciones mediante signos diferentes, los numerales de un color eran utilizados para expresar los valores positivos y los de otro color para indicar los negativos ... o que la notaci6n negra era para sumar y la blanca para restar // because of the necessity to represent operations through different signs, the numerals of one color were utilized to express positive values and the other color negative values ... or that the black notation was to add and the white to subtract" (Primeglio 33).

This seems out of place, given that the 55 place settings on the table correspond to pendant cords on the displayed *quipo*. There should not be a notion of negative pendant cords. Computations were simplified without a notation of negative numbers, as shown in extant grave *quipos*.

Guaman Poma's other *quipo* pages shed light on this problem. Building on Zuidema's idea, I examined Guaman Poma's proclivity to "agrupar listas de personajes, etapas cronol6gicas y otras formas numerativas // group lists of personalities, chronological eras, and other countable subjects" (Ossio 47). In two other examples of depiction of administrators and their *quipos*, the numbers speak for themselves. Here again on page 335 and 348, the text matches the illustrations of *quipos* (Plates 5 and 6). In the depiction on page 335, is a drawing of the presentation of a *quipo* to Topa Inca Yupanqui, for deposits of the goods, in a *collca*. Here Guaman Poma enters a list of goods and products normally stored in these warehouses. A match between the numbers of items listed can be easily read as a multiple of the actual *quipo* pendant cords represented in the image.[6] Twelve goods are listed in the text (*chuno, lana, axi, coca*, etc.) There appear to be 24 pendant cords on the illustrated *quipo*. To understand this "doubling" connection between *quipo* and *collca*, one must consider dual *ayllu* organization of Andean polities; whereby all Andean society was divided in dual terms and relationships. I contend that the dual division in the representation of the storehouses on page 335 of the *Nueva Cor6nica*, on the bottom and top on the page, represents the doubling of items of this particular *quipo*.

The next drawing on page 348, without a structural or architectural context, is more akin to the representation of the *quipo* on page 360. Here Guaman Poma depicts a provincial administrator with two *quipos*. The text enumerates two lists this time, and

---

[6] John Murra also discusses a colonial list of goods and argues that it is based on a direct *quipo* transcription.

here the match between *quipo* pendant cords and lists are perfect. There are seven and 21 items in each list and these match exactly the pendant cords on the illustrated *quipos*. These correlations provide strong evidence for the mnemonic association between pendant cords and particular objects. This association should also prove, in regards to the table on page 360, that a black dot on the table does not necessary denote zero. The difference between placeholders is clearly not black and white when referring to the counting table of the Nueva Cor6nica. The fact that knots and cords had different colors, revises Wassen's binary opposition. A more fruitful reading would suggest an iconographic reading of *quipos* to include both form (knots, ties, and twists) and different colors.

Toward iconographic reading of *quipos*: Inca dress and color

Scholarship may draw new models for *quipos*, mnemonic or numeric, meaning by using current arguments used in analysis of royal Inca dress and weaving patterns. Both Cummins and Zuidema suggest that Inca dress, and the colors associated with these weaves, signify on levels far beyond mere patterns and numbers. Cummins writes patterns "reproduced forms corresponding to known world; they were not mimetic. They were abstractions based on color and formal arrangement, like the *quipos*, and they revealed their content through the interpretation of informed individuals" (99). Whereas dress has received ample analysis, *quipos* have not been addressed using this iconographic methodology. The problems of past scholarship are due in part to the lack of adequate knowledge about color schemes in extant *quipos*. Yet, even for dress, iconographic evidence is fragmented. "*Tucapus*-square and mostly abstract signs used on Inca textiles and in other media were part of a complicated system of graphic communication, but no chronicler ever represented on it directly" (Zuidema 151). Given the ability to convey complex information iconographically, dress and *quipos* could fall under the same descriptive structure. Zuidema also mentions the connection between *collca* storehouses and what are termed *collca* motifs in royal dress (167). These designs belonged to the same hierarchical structure, along with *quipo*, on which other forms of imperial organization were built. According, to certain descriptions, these tunics are characterized by the repetitive use of certain motifs and provided information about the "social, ritual, and mythical, temporal, and the even effective role of these designs" (Zuidema 152; Plate 7).

Toward this iconographic reading, we might apply some ideas generated by the text found in the last page of Friar Martin de Murua's *Historia General de Peru*, in the Loyola manuscript (Plate 8). He presents "the structure of the weave of *chumbi*, a belt, and the *Quecha* prayers that maybe related to it, perhaps in the way that the *quipos* structure and color are related to the data it represents" (Cummins 197). The weave reveals color, knot associations that were very important to *quipo* reading. From other drawings present in the known three versions of Murua's manuscript, the evidence is suggestive of the same color connections. *Quipos* are always represented as different colored ropes with successive ties and pendant cords. Murua' s written account also states that knots were supposed to indicate figures and the color of the cords indicated subject matter. "A conventional color scheme was in use, with certain colors for different kinds of things to be recorded" (Locke 3). Other evidence suggests that *quipos* were systematized, so that accurate records could be read across the empire. It would make

sense that if color codes were not systematized, they at least referred to common motifs and colors, much like examples in Inca dress.

Other work by Gary Urton suggests schematic color use on extant *quipos*. He has done much work to advance *quipo* understanding through grave *quipos*[7] and *quipos* belonging to colonial *quipocamayocs*. This recent research elucidates a type of noun and number correlations in the reading of a specific *quipo* belonging to two caciques from the region of the northeastern Andes of Peru (Urton 418 and; in 2001). The *quipo* described is one of 32 *quipos* discovered, along with some 220 mummy bundles, in 1996 in a half-dozen *ckullpas* (burial houses) built into a rock-overhang overlooking a lake, called Laguna de los C6ndores, near the town of Leymebamba. The cultural materials found with the mummies and *quipos* date from the pre-Inkaic Chachapoya culture (ca. A.D. 800--1450), through the Inka occupation of the region and on into the early colonial era. It is argued that the *quipos* stored with the dead represented tomb texts, which contained information pertaining to the history of the mummies and the social groups descended from them.

One of the *quipo* samples (UR6) is interpreted as a combined biennial calendar and census of the tribute payers. It is argued that *quipo* UR6 was the source of information for the first colonial Spanish census in the region, in 1535. Urton asserts that ingrained in all *quipos* are at least some notions of nouns, which identified tribute items, and the numbers, which were used in calculating by means of stones the total number of units of each item. Could these stones have some relation to Guaman Poma's counting table? Urton's work suggests such techniques could provide a systematized counting system throughout the Inca Empire. A system could be identified with "some numerical value that served as a power, or multiplier, to be used in calculating" quantities in question (Urton 418). Guaman Poma's table could be evidence of such a system, in which multipliers fit into columns and rows, to systematize the knotting of a *quipo* with different colors and numerical quantities.

Guaman Poma unfortunately did not paint his illustrations in color in the *Nueva Coronica*. His only mention to color and its use in *quipos* can be found on page 359, and like his other descriptions of lnca dress, only announces that *quipos* "colores tenido // I had color." On the other hand Muma does have depictions of colored *quipos*, specifically in a newly discovered Irish manuscript. Not only are pendant cords painted with different colors, as Ossio believes the same artistic hand may have been used in their production, possibly that of Guaman Poma.

> "Es cierto que no son identicas pero, en primer lugar, los colores de los vestidos son exactamente los mismos que los referidos para cada uno de estos personajes en el text y en los mismos dibujos de la *Nueva Coronica* // It is true that the drawings are not identical, but the colors of the clothes are exactly the same to those referred to for each personalities in the text and the drawing of the *Nueva Coronica*" (Ossio 49).

Ossio suggest that this co-authorship of the Murua manuscript with Guman Poma may explain why we find color in the manuscript *quipo* drawings. Yet, no color pattern is

---

[7] Gary Urton "Messages from the Mallquis," unpublished paper.

readably discernable, because unlike *quipo* counterparts in the *Nueva Coronica*, these *quipos* are not supported with ample text. Often it is in this association, between text and image, that one can derive intra-textual meaning. Murua does write though

> "y para que las cosas que querlan contar diferenciasen, hacen los nudos mayores y menores y con diferencia de colores, de manera que para una cosa tenian nudo colorado, y para otra, verde o amarillo, y as! iba lo demas // and to differentiate the things being counted, they made the major and minor knots with different colors, in a manner that for one thing they had a red knot, and for another, green or yellow, and so on" (225).

We can gather then, just as pendant cords denote subject, each pendant was color-coded per subject matter. Still, for interpreting specific meaning, there is no simple model. The only correlations presented are those found in grave *quipos*, and their connections with magic numbers. "With regard to different colours, I believe they were employed merely to make possible several combinations of numbers which would include seven, and occasionally even astrological numbers"(Nordenskiold 14). No structure proposed by any sources, either textual or illustrations, can serve to make numerology a possible explanation for all *quipo*. The idea that these numbers were plainly magic is not enough to understand the subtle differences in *quipo* codes. There are more than just numbers at work. *Quipo* iconography and *tucapu* dress were universally transferable across the empire. Therefore, *quipo* code must have been enumerated and standard.

Quipo **and Cryptography**

Where efforts to interpret *quipo* code are concerned, attention should be directed toward more analysis of form, rather than function. More attention should be addressed to the actual use of color. Much like Inca dress, and the patterns they used as references, color clues of this type can be found in Guaman Poma. In the table illustrated on page 360, we find the nearest example of the many mnemonic levels and the color elements that make a reading of *quipos* standard and possible throughout the Inca Empire. Taking color schematics into account, the table can no longer suggest just a simple abacus. First, the table, like that one described by Father Acosta, is not a plausible counting device. Along with my own list analysis presented earlier, there are other problems still to be addressed in Guaman Poma's table. Taking the table and its 55 placeholders as a starting point, the very nature of the table changes from a simple abacus, as suggested by Wassen, to a mnemonic device, indicated in the previous color analysis. Guaman Poma on the same page writes that *quipocamayos* "Cuentan en tables// They count in tables" (361). Tabla here was referential word for lists, with its Spanish connotation. Did the Inca count in lists? It is more likely that the Inca would read in *quipos* and count on tables.

There has been no analysis since Wassen to account for the series of numbers on the horizontal axis, without linking them to magic numbers. I would like to suggest that this very table, and its horizontal axis, maybe key to deciphering the Inca *quipo* code. The table could have been a system by which *quipos* might have been systematized across the empire. First, it is not a square, but a rectangle, an unconventional way to build an abacus. Second the placement of white and black on the table vary and are not systematic. They have no inherent structure apart from specific placement and

numeration in each localized square. Color might have been an extra variable that would allow us to examine the use of black dots and white dots further. Third, for all the scholarship surrounding this rectangle, no one has picked up on the sequence of numbers present on the horizontal axis. The numbers 1, 2, 3, 5 are a classic Fibonacci sequence with the formula:

$$f(l) + f(2)+f(3)+ ..... f(n)= f(n+2)-1$$

This sequence is known and studied because of its frequent occurrence in scientific models of nature. If we apply this sequence to the table, we would gather that the Incas, used base ten rather than binary mathematics, which from previous is known to be true.[8]

Modern uses for the Fibonacci sequence numbers include cryptography. If so, a pattern would be logically present in the drawing of the Guaman Poma table, for the use of systematized Inca code to write and read their *quipos*. Variables may still exist for double coding, because of the use of white and black. These variables could multiply if we assume that other colors were involved in table computation. *Quipocamayocs* could have written their codes in binary or in more numerals for additional colors. Approximating a system to show meaning through color analysis would be difficult. Guaman Poma's table could begin to look more like a simple computer, like loom cards invented in 18th century Europe. Theoretical computer science might shed some light on such structures present in Guaman Poma's table.

In review, it is clear, that in Guaman Poma's case, *quipo* writing is not like phonetized writing or even akin to traditional mnemonic devices. It is much more like scientific notation, or somewhat like musical notation where symbols hold meaning per their relationship to each other and the physical world they try to express. It should be no surprise that most work done on *quipos* has been conducted from a mathematical analysis perspective. It also should be clear that this analysis tackles very few of the symbolic problems involved in acquiring meaning from a *quipo*. Scholarship should start from an iconic frame of reference and ground itself in what is known of Inca society. From this basis, the question remains less about mathematics and more about symbolism. These gray areas are often the language markers with links to codified symbolic forms. Understanding this mechanism of reference and symbols may in turn speak to the very formation of our own ideas of what can be represented. The Inca example of *quipos* is interesting in that it shows a concerted effort on the part of a culture to codify knowledge through color and number, not words.

Bibliography

Adorno, Rolena

1989. Cronista y Principe. Lima: Fondo Editorial de la Ponitifica Universidad Catolica de! Peru

1991. Images of Indios Ladinos in Early Colonial Peru. in Transatlantic Encounters. Ed. Kenneth Andrien. University of California Press.

Ascher, Maria and Robert Ascher

1981. Code of the Quipo. Ann Arbor: University of Michigan Press

Cummings, Tom

[8] Ascher and Ascher. Quipo Code. Ann Arbor: Univesity of Michigan Press, 1981.

1994. Colonial Image of the Inca. in Writing Without Words. Ed. Elizabeth Boone. Durham: Duke University Press.

Day, Cyrus Lawrence

1967. Quipos and Witches Knots. Lawrence: University of Kansas Press.

Derrida, Jacques

1976. Of Grammatology. Trans. Gayatri Spivak Baltimore: The John Hopkins University Press.

Guaman Poma de Ayala, Felipe

1968. El Primier Nueva Cronica y Buen Gobierno. Paris:Institut d'Ethnologie.

Locke, Leland

1927. A Peruvian Quipu. Contributions from the Museum of the American Indian Heye Foundation, VII, No.5.

Markham, Clements

1920. Introduction. in Memorias antiguas de! Peru, by Fernando Montesinos. Trans. And Ed.

Philip A.Means. London:The Hakluyt Society. 2"d ser. No. XL VIII, pg. 3-15.

Molina, Christova de

1873. The fables and rites of the Yncas. in Rites and Laws of the Y ncas. Trans. and Ed. Clements Markam. London: The Hakluyt Society. pg.1-64.

Murra, John V. and Craig Morris

1976. Dynasty Oral Tradition, Administrative Records and Archaeology in the Andes. in World Archaeology, vol.7, no.3, pp269-279.

Murua, Fr. Matrin de

1963. Histroria General de! Peru. Torno I. Madrid: Instituto Gonzalez Fernandez de Oviedo.

Nordenskiold, Briand

1925. The secret of the Peruvian Quipos. Goteberg: Comparative Ethnographical Studies, VI.

Ossio, Juan M.

2000. Guaman Poma y Murua: ante la tradicion oral andina. in Iconos. No. 4, Peru.

Prescott, William H.

1968. History of the conquest of Peru. [1847]. New York: AMS Press, 2vols.

Primeglio, Carlos Radicatti di

1980. Sistema Contable de los Incas. Lima: Liberia Studium esa.

Rowe, John H ..

1945. Absolute chronology in the Andean area. in American Antiquity. Vol. X, no. 3, pp.265-284.

1946. Inca culture at the Time of Spanish Conquest. in Handbook of South American Indians. Ed. Julian H. Steward.Washington: Bureau of American Ethnology Bulletin 143, pp.183-330.

Wassen, Henry

1931. The Ancient Peruvian Abacus. Goteberg: Comparative Ethnographical Studies, IX. U11on, Gary

1998. From Knots to Narratives: Reconstructing the Art of Historical Record Keeping in the Andesfi·om Spanish Transcription to lnka Khipus. Ethnohistory.

V.45 no.3 2001 A Calendrical and Demographic Tomb Text. in Latin American Antiquity Vol. 12, Number 2, June.

Zuidema, R.Tom

1991. Guaman Poma and the Art of Empire: Toward an Iconography of Inca Royal Dress. In Transatlantic Encounters. Ed. Kenneth Andrien. University of California Press.